

SOLUTIONS

MODEL QUESTION PAPER

PHYSICS (Class XII) – Chapter 4
Moving Charges and Magnetism

This is a full, worked solution to every question in the companion model question paper (all 29 questions, not only the ones you'd attempt in the exam), so that you can check the exact method and steps expected by the board, not just the final answer. Marks in brackets indicate the weight of each part.

Part I Very Short Answer (1 mark each)

1. Define the SI unit of magnetic field, the tesla, in terms of the force on a moving charge.

Solution. From $F = qvB \sin \theta$, if $q = 1 \text{ C}$, $v = 1 \text{ m/s}$, $\theta = 90^\circ$ and $F = 1 \text{ N}$, then $B = 1 \text{ T}$.
1 tesla is the magnetic field in which a charge of 1 coulomb, moving with a velocity of 1 m/s perpendicular to the field, experiences a force of 1 newton.

2. State the rule used to find the direction of the force on a positive charge moving in a magnetic field.

Solution. The **right-hand (screw) rule**, applied to the vector product $\vec{F} = q(\vec{v} \times \vec{B})$: point the fingers along $\vec{v}$, curl them towards $\vec{B}$; the thumb gives the direction of $\vec{F}$ (for a positive charge).

3. Fill in the blank: To convert a moving-coil galvanometer into an ammeter, a resistance of _____ value is connected in _____ with the coil.

Solution. **Low** value; connected in **parallel** (as a shunt) with the coil.

4. What is the nature of the path of a charged particle moving at an angle of 30° to a uniform magnetic field?

Solution. Helical. The velocity component along $\vec{B}$ ($v \cos 30^\circ$) is unaffected and gives uniform translation; the component perpendicular to $\vec{B}$ ($v \sin 30^\circ$) produces circular motion. The two combined give a helical path.

5. Write the SI unit of magnetic moment.

Solution. Ampere metre squared, **A m²** (equivalently, joule per tesla, J/T).

6. State whether True or False: “Two magnetic field lines can intersect each other.”

Solution. False. At the point of intersection the field would have two different directions simultaneously, which is not possible for a single-valued vector field.

7. Name the device that uses crossed electric and magnetic fields to select charged particles of a specific velocity.

Solution. Velocity selector. It works on the principle that the particle moves undeflected only when the electric force qE exactly balances the magnetic force qvB , i.e. when $v = E/B$.

Part II Short Answer – I (2 marks each)

- Write down the expression for the Lorentz force on a charge q moving with velocity $\vec{v}$ in the presence of both an electric field $\vec{E}$ and a magnetic field $\vec{B}$. Explain why the magnetic part of this force does no work on the charge.

Solution. $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}) = \vec{F}_{\text{electric}} + \vec{F}_{\text{magnetic}}$

The magnetic part $\vec{F}_{\text{mag}} = q(\vec{v} \times \vec{B})$ is, by the property of the vector cross product, always perpendicular to $\vec{v}$. Work done in a small time dt is $dW = \vec{F}_{\text{mag}} \cdot \vec{v} dt$. Since $\vec{F}_{\text{mag}} \perp \vec{v}$, $\vec{F}_{\text{mag}} \cdot \vec{v} = 0$, so $dW = 0$. Hence the magnetic force can change only the *direction* of velocity, never its magnitude (speed/KE).

- State Ampere's circuital law and write its mathematical form.

Solution. Ampere's circuital law states that the line integral of the magnetic field $\vec{B}$ around any closed loop is equal to μ_0 times the total steady current enclosed by that loop.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enc}}$$

- A proton and an alpha particle enter a uniform magnetic field with the same speed, moving perpendicular to $\vec{B}$. Given that the mass and charge of the alpha particle are 4 and 2 times that of the proton respectively, compare the radii of their circular paths.

Solution. $r = \frac{mv}{qB}$, with v and B the same for both particles.

$$\frac{r_\alpha}{r_p} = \frac{m_\alpha}{m_p} \cdot \frac{q_p}{q_\alpha} = \frac{4m_p}{m_p} \cdot \frac{q_p}{2q_p} = \frac{4}{2} = 2$$

So $r_\alpha : r_p = 2 : 1$ — the alpha particle's circular path has twice the radius of the proton's.

- Distinguish between the current sensitivity and the voltage sensitivity of a moving-coil galvanometer.

Solution. Current sensitivity is the deflection produced per unit current: $\frac{\phi}{I} = \frac{NAB}{k}$.

Voltage sensitivity is the deflection produced per unit voltage: $\frac{\phi}{V} = \frac{NAB}{kR}$, where R is the coil resistance.

Increasing the number of turns N always increases current sensitivity, but since R is also proportional to N (more wire), voltage sensitivity may remain unchanged even though current sensitivity increases.

- State the rule used to determine the direction of the magnetic field due to (a) a long straight current-carrying conductor, and (b) a circular current loop.

Solution. (a) **Right-hand rule:** grasp the wire with the right hand, thumb pointing along the current; the curled fingers give the direction of the circular field lines around the wire.
 (b) **Right-hand thumb rule:** curl the fingers of the right hand along the direction of current flow in the loop; the extended thumb gives the direction of $\vec{B}$ on the axis.

6. Two long, straight, parallel conductors carry currents in the same direction. Using the expression for the force per unit length between them, explain why they attract each other.

Solution. Force per unit length: $f = \frac{\mu_0 I_a I_b}{2\pi d}$.

The field B_a produced by conductor 'a' at the location of 'b' is found using the right-hand rule; for currents in the same direction, this field is such that the force $\vec{F} = I_b(\vec{l} \times \vec{B}_a)$ on 'b' points *towards* 'a'. By Newton's third law, 'a' experiences an equal force towards 'b'. Hence the two parallel, same-direction currents pull towards each other – they attract.

7. Define the ampere, the SI unit of electric current, using the force between two current-carrying conductors.

Solution. Setting $I_a = I_b = 1$ A and $d = 1$ m in $f = \frac{\mu_0 I_a I_b}{2\pi d}$:

$$f = \frac{4\pi \times 10^{-7}}{2\pi} = 2 \times 10^{-7} \text{ N/m}$$

One ampere is defined as that constant current which, if maintained in each of two infinitely long, straight, parallel conductors of negligible cross-section placed one metre apart in vacuum, produces a force of 2×10^{-7} newton per metre of length between them.

Part III Short Answer – II (3 marks each)

1. A charged particle q moving with velocity $\vec{v}$ experiences a force in the presence of both an electric field $\vec{E}$ and a magnetic field $\vec{B}$. [Previous Year Question – Say 2011, adapted]
 - (a) Write down the expression for this force. (1)
 - (b) A configuration of mutually perpendicular electric and magnetic fields, used to select charged particles of a particular speed, is called a _____. (1)
 - (c) Name the cyclic accelerator that uses a magnetic field to accelerate charged particles to high energies. (1)

Solution. (a) $\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$ (b) **Velocity selector** (c) **Cyclotron**

2. Derive an expression for the radius of the circular path described by a charged particle moving with speed v , perpendicular to a uniform magnetic field B . Hence obtain an expression for its cyclotron frequency, and state one important property of this frequency.

Solution. When $\vec{v} \perp \vec{B}$, the magnetic force qvB acts perpendicular to $\vec{v}$ at every instant and therefore serves as the centripetal force for circular motion:

$$\frac{mv^2}{r} = qvB \Rightarrow r = \frac{mv}{qB}$$

Since $v = \omega r$, the angular frequency is

$$\omega = \frac{qB}{m} \Rightarrow \nu = \frac{\omega}{2\pi} = \frac{qB}{2\pi m} \quad (\text{cyclotron frequency})$$

Property: ν is independent of the particle's speed v (and hence of its radius r or kinetic energy) – it depends only on q , B , and m . This is exactly what allows a cyclotron to accelerate a particle using an oscillating voltage of one fixed frequency, even as the particle's speed keeps increasing.

3. State the Biot–Savart law in vector form. Mention **two** points of similarity and **two** points of difference between the Biot–Savart law for a magnetic field and Coulomb's law for an electric field.

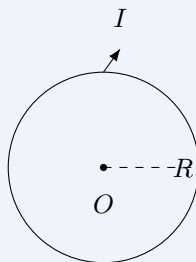
Solution.

$$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{r^2}$$

Similarities: (i) Both are long-range, inverse-square laws ($\propto 1/r^2$). (ii) Both obey the principle of superposition.

Differences: (i) Coulomb's law arises from a scalar source (charge q); the Biot–Savart law arises from a vector source (current element $I d\vec{l}$). (ii) The electrostatic field is directed along the displacement vector $\vec{r}$; the magnetic field is perpendicular to the plane containing $d\vec{l}$ and $\vec{r}$. (There is also an explicit $\sin \theta$ angle-dependence in the Biot–Savart law that has no counterpart in Coulomb's law.)

4. Derive an expression for the magnetic field at the centre of a circular coil of N turns and radius R , carrying a steady current I .

Solution.

Consider a circular loop of radius R carrying current I . Every element $I d\vec{l}$ of the loop is at the same distance R from the centre O , and $d\vec{l} \perp \hat{r}$ everywhere ($\theta = 90^\circ$), so by the Biot-Savart law each element contributes

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin 90^\circ}{R^2} = \frac{\mu_0 I dl}{4\pi R^2}$$

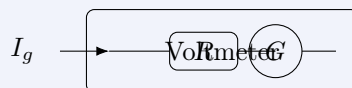
All such contributions point in the *same* direction (perpendicular to the plane of the loop, by the right-hand rule), so they simply add:

$$B = \int dB = \frac{\mu_0 I}{4\pi R^2} \int dl = \frac{\mu_0 I}{4\pi R^2} (2\pi R) = \frac{\mu_0 I}{2R}$$

For N turns, each contributes equally, so

$$B = \frac{\mu_0 N I}{2R}$$

5. With the help of a labelled circuit diagram, explain how a moving-coil galvanometer of resistance G and full-scale current I_g can be converted into a voltmeter of range 0 to V volts. Obtain the expression for the resistance to be used.

Solution.

A large resistance R is connected in **series** with the galvanometer. When the maximum voltage V to be measured is applied across this series combination, the current through it must be exactly the full-scale current I_g :

$$V = I_g(R + G) \Rightarrow R = \frac{V}{I_g} - G$$

Since R is large, the total resistance $R + G \approx R$ is large, so the voltmeter draws only a very small current from the circuit and does not appreciably disturb the voltage being measured.

6. A straight wire carrying a current I is bent into a semicircular arc of radius R .
- (a) What is the magnetic field at the centre of the arc due to the straight parts? (1)
- (b) Derive the expression for the magnetic field at the centre due to the semicircular part. (2)

Solution. (a) **Zero.** For the straight parts, $d\vec{l}$ is directed radially towards/away from the centre, so $d\vec{l} \parallel \hat{r}$ and $d\vec{l} \times \hat{r} = 0$; hence $dB = 0$ everywhere along the straight segments.

(b) For the semicircular part, every element $I dl = IR d\theta$ is at distance R from the centre with $d\vec{l} \perp \hat{r}$, so

$$dB = \frac{\mu_0}{4\pi} \frac{I R d\theta}{R^2} = \frac{\mu_0 I}{4\pi R} d\theta$$

All contributions point in the same direction, so integrating θ from 0 to π :

$$B = \int_0^\pi \frac{\mu_0 I}{4\pi R} d\theta = \frac{\mu_0 I}{4\pi R} \times \pi = \boxed{\frac{\mu_0 I}{4R}}$$

(This is exactly half the field of a full circular loop, $\mu_0 I/2R$, as expected since a semicircle is half a loop.)

7. A rectangular current loop is placed in a uniform magnetic field $\vec{B}$. Show that the loop experiences a torque but no net force. Write the expression for this torque in terms of the magnetic moment $\vec{m}$ of the loop.

Solution. Let the loop $ABCD$ have sides a ($=BC=AD$) and b ($=AB=CD$), current I , with the normal to the loop making angle θ with $\vec{B}$.

Forces on BC and AD are equal, opposite, and collinear (along the loop's axis) – they cancel exactly, contributing no net force and no torque.

Forces on AB and CD have equal magnitude $F_1 = F_2 = IbB$ but act in opposite directions and are *not* collinear (separated by a perpendicular distance $a \sin \theta$) – so while the net force is still zero, they form a couple:

$$\tau = F_1 \cdot \frac{a}{2} \sin \theta + F_2 \cdot \frac{a}{2} \sin \theta = IbB a \sin \theta = I(ab)B \sin \theta = IAB \sin \theta$$

In vector form, defining the magnetic moment $\vec{m} = I\vec{A}$ (direction by the right-hand rule):

$$\boxed{\vec{\tau} = \vec{m} \times \vec{B}}$$

So the loop experiences **zero net force** but a **non-zero torque** (except when $\theta = 0$ or 180°).

Part IV Long Answer – I: Numerical Application (4 marks each)

1. A charged particle of mass 1×10^{-26} kg and charge 1.6×10^{-19} C, travelling with velocity 1.28×10^6 m/s along $+x$, enters a region where a uniform electric field ($E_z = -1.024 \times 10^5$ V/m) and a uniform magnetic field ($B_y = 8 \times 10^{-2}$ T) exist together. [Previous Year Question – March 2009, adapted]
 - (a) Write the expression for the force on the particle due to the electric field alone. (1)
 - (b) Calculate the magnitude and direction of this electric force. (2)
 - (c) Write the expression for the force experienced by the particle due to the magnetic field alone. (1)

Solution. (a) $\vec{F}_E = q\vec{E}$

(b) $F_E = qE_z = (1.6 \times 10^{-19})(1.024 \times 10^5) = 1.638 \times 10^{-14}$ N. Since q is positive and E_z is along $-z$, the force is also along $-z$:

$$F_E \approx 1.64 \times 10^{-14} \text{ N, along } -z$$

(c) $\vec{F}_B = q(\vec{v} \times \vec{B})$

2. A circular coil of wire consisting of 100 turns, each of radius 8.0 cm, carries a current of 0.40 A.
 - (a) Write the formula for the magnetic field at the centre of a coil of N turns. (1)
 - (b) Calculate the magnitude of the magnetic field B at the centre of the coil. (3)

Solution. (a) $B = \frac{\mu_0 NI}{2R}$

(b) Given $N = 100$, $I = 0.40$ A, $R = 8.0$ cm $= 8 \times 10^{-2}$ m:

$$B = \frac{(4\pi \times 10^{-7})(100)(0.40)}{2(8 \times 10^{-2})} = \frac{4\pi \times 10^{-7} \times 40}{0.16}$$

$$B \approx 3.14 \times 10^{-4} \text{ T}$$

3. A closely wound solenoid, 80 cm long, has 5 layers of windings of 400 turns each. The diameter of the solenoid is 1.8 cm and it carries a current of 8.0 A.
 - (a) Write the formula for the magnetic field inside a long solenoid. (1)
 - (b) Estimate the magnitude of B inside the solenoid, near its centre. (3)

Solution. (a) $B = \mu_0 nI$, where n = number of turns per unit length.

(b) Total turns $= 5 \times 400 = 2000$. Length $l = 0.80$ m.

$$n = \frac{2000}{0.80} = 2500 \text{ turns/m}$$

$$B = \mu_0 nI = (4\pi \times 10^{-7})(2500)(8.0)$$

$$B \approx 2.51 \times 10^{-2} \text{ T}$$

(The diameter is not needed for this estimate – it only matters for how “long” the solenoid is considered relative to its radius.)

4. A moving-coil galvanometer has a coil resistance of 12Ω and gives full-scale deflection for a current of 3 mA.

- (a) How should it be converted into an ammeter of range 0–3 A? Find the value of the shunt resistance required. (2)
- (b) How should it be converted into a voltmeter of range 0–15 V? Find the value of the series resistance required. (2)

Solution. Given $G = 12 \, \Omega$, $I_g = 3 \times 10^{-3} \, \text{A}$.

(a) Connect a small shunt resistance S in **parallel** with the coil, so that of the total current $I = 3 \, \text{A}$, only I_g passes through the galvanometer and the rest through the shunt:

$$S = \frac{I_g G}{I - I_g} = \frac{(3 \times 10^{-3})(12)}{3 - 3 \times 10^{-3}} = \frac{0.036}{2.997}$$

$$\boxed{S \approx 1.20 \times 10^{-2} \, \Omega}$$

(b) Connect a large resistance R in **series** with the coil:

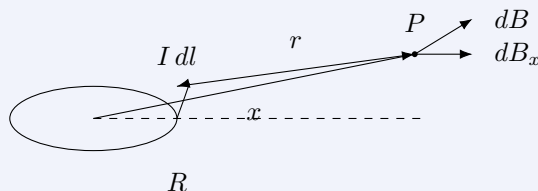
$$R = \frac{V}{I_g} - G = \frac{15}{3 \times 10^{-3}} - 12 = 5000 - 12$$

$$\boxed{R = 4988 \, \Omega}$$

Part V Long Answer – II: Derivations (5 marks each)

1. Using the Biot–Savart law, derive an expression for the magnetic field at a point on the axis of a circular current loop of radius R carrying current I , at a distance x from its centre. Hence obtain the magnetic field at the centre of the loop.

Solution.



Consider a circular loop of radius R , carrying current I , in the y - z plane with centre O . Let P be a point on the axis (the x -axis) at distance x from O . Consider a current element $I d\vec{l}$ at the top of the loop. Its distance from P is $r = \sqrt{x^2 + R^2}$, and since $d\vec{l} \perp \vec{r}$ everywhere on this axis,

$$dB = \frac{\mu_0}{4\pi} \frac{I dl \sin 90^\circ}{r^2} = \frac{\mu_0}{4\pi} \frac{I dl}{x^2 + R^2}$$

This dB is perpendicular to $\vec{r}$ and can be resolved into a component dB_x along the axis and $dB_\perp$ perpendicular to it. The $dB_\perp$ components from diametrically opposite elements cancel by symmetry, leaving only the axial components, with $\cos \phi = R/\sqrt{x^2 + R^2}$:

$$dB_x = dB \cos \phi = \frac{\mu_0}{4\pi} \frac{I dl}{x^2 + R^2} \cdot \frac{R}{\sqrt{x^2 + R^2}}$$

Integrating around the full loop ($\int dl = 2\pi R$):

$$B = B_x = \frac{\mu_0}{4\pi} \frac{IR}{(x^2 + R^2)^{3/2}} (2\pi R)$$

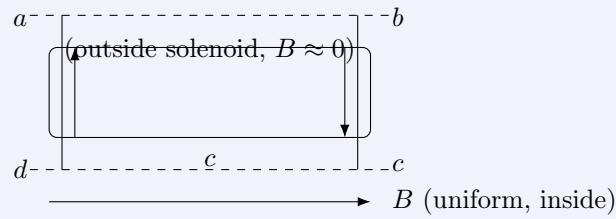
$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

At the centre ($x = 0$):

$$B = \frac{\mu_0 I}{2R}$$

2. State Ampere's circuital law. Using it, derive an expression for the magnetic field (a) inside, and (b) outside a long, straight, current-carrying solenoid having n turns per unit length and carrying current I .

Solution. Ampere's circuital law: $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$, the line integral of $\vec{B}$ around a closed loop equals μ_0 times the current enclosed.



Consider a rectangular Amperian loop $abcd$, with side ab (length l) inside the solenoid and parallel to the axis, and cd outside the solenoid.

(a) Inside the solenoid: Along ab , $\vec{B}$ is uniform, strong, and parallel to $d\vec{l}$, contributing Bl . Along the two transverse sides bc and da , $\vec{B} \perp d\vec{l}$, so these contribute zero. Along cd (outside), $B \approx 0$ for a long solenoid, so this side contributes zero too. Hence

$$\oint \vec{B} \cdot d\vec{l} = Bl$$

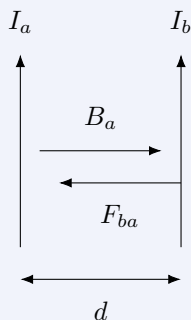
The current enclosed by this loop is $I_{enc} = (nl)I$, since nl turns pass through the loop. By Ampere's law:

$$Bl = \mu_0(nl)I \quad \Rightarrow \quad \boxed{B = \mu_0 n I \quad (\text{inside, parallel to axis})}$$

(b) Outside the solenoid: For an ideal long solenoid, the field lines that spread out are negligibly weak (they must return over a very large area), so

$$\boxed{B \approx 0 \quad (\text{outside})}$$

3. Derive an expression for the force per unit length between two long, straight, parallel current-carrying conductors placed a distance d apart. Hence define the SI unit of current, the ampere.

Solution.

Let two long, straight, parallel conductors a and b , separated by distance d , carry currents I_a and I_b in the same direction.

The field produced by conductor a at the location of conductor b is

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

(directed as given by the right-hand rule, e.g. downward if the currents point upward).

The force experienced by a length L of conductor b , carrying current I_b in this field, is

$$F_{ba} = I_b L B_a = I_b L \left(\frac{\mu_0 I_a}{2\pi d} \right) = \frac{\mu_0 I_a I_b L}{2\pi d}$$

directed towards a (an attractive force, for currents in the same direction). Hence the **force per unit length** is

$$f = \frac{F_{ba}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$$

By Newton's third law, conductor a experiences an equal and opposite force per unit length due to b . (Antiparallel currents repel, by the same reasoning with the field/force directions reversed.)

Definition of the ampere: Setting $I_a = I_b = 1$ A and $d = 1$ m:

$$f = \frac{\mu_0}{2\pi} = \frac{4\pi \times 10^{-7}}{2\pi} = 2 \times 10^{-7} \text{ N/m}$$

One ampere is the constant current which, maintained in each of two infinitely long, straight, parallel conductors of negligible cross-section, placed one metre apart in vacuum, would produce a force of 2×10^{-7} N per metre of length on each conductor.

4. A rectangular loop of area A , carrying a steady current I , is placed in a uniform magnetic field $\vec{B}$ such that the normal to the loop makes an angle θ with $\vec{B}$.
 - (a) Derive an expression for the torque experienced by the loop. (3)
 - (b) Briefly describe how this principle is used in the construction and working of a moving-coil galvanometer. (2)

Solution. (a) Let the rectangular loop $ABCD$ have sides a and b (area $A = ab$), carrying current I , with its normal (area vector $\vec{A}$) making angle θ with $\vec{B}$.

The forces on sides BC and AD are equal, opposite, and collinear along the loop's rotation axis – they cancel with no net torque.

The forces on AB and CD have equal magnitude

$$F_1 = F_2 = IbB$$

act in opposite directions, and are separated by a perpendicular distance $a \sin \theta$ (since the loop is tilted by θ from the field direction), forming a couple:

$$\tau = F_1 \left(\frac{a}{2} \sin \theta \right) + F_2 \left(\frac{a}{2} \sin \theta \right) = IbB a \sin \theta$$

$$\tau = IAB \sin \theta = \vec{m} \times \vec{B}, \quad \text{where } m = IA$$

For N turns, $\tau = NIAB \sin \theta$, with $\vec{m} = NI\vec{A}$.

(b) **Moving-coil galvanometer:** A rectangular coil of N turns and area A is suspended in a *radial* magnetic field B (produced using curved pole pieces and a soft-iron core), so that the plane of the coil is always parallel to B and $\sin \theta = 1$ at every position – giving a constant torque $\tau = NIAB$ for a given current I , independent of the coil's orientation. This magnetic torque twists a restoring spring, which provides a counter-torque $k\phi$ (k = torsional constant, ϕ = deflection). At equilibrium,

$$NIAB = k\phi \quad \Rightarrow \quad \phi = \left(\frac{NAB}{k} \right) I$$

Since the quantity in brackets is a constant for a given instrument, $\phi \propto I$: the deflection on the linear scale directly indicates the current, which is the working principle of the galvanometer.